LESSON 10.8
SOLVING REAL-WORLD PROBLEMS WITH TWO-STEP EQUATIONS

LESSON INTRODUCTION

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.

LEARNING TARGETS

- I can solve two-step equations within a real-world context.
- I can relate the solution of an equation representing a real-world context to the context.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

WORKING TOWARD

MA.8.AR.2.1 Solve multi-step linear equations in one variable with rational number coefficients. Include equations with variables on both sides.

ESSENTIAL QUESTIONS

- How do you relate the solution of an equation representing a real-world context to the context?
- How do you solve two-step equations?

LESSON NARRATIVE

This is the first of three lessons in which students are guided through writing equations in the form $px + q = r$ and $p(x + q) = r$. The lesson is scaffolded throughout to help build student thinking about real-world contexts and the equations that model them. Students also relate given equations to the real-world contexts they represent. Finally, students solve and interpret solutions based on the real-world context.

COMPONENT SUMMARY

10.8.1 WARM-UP *(~5 MINUTES)*

Describe a pattern

10.8.2 COLLABORATION *(10 MIN.)*

Match equation to bar diagrams and real-world contexts

10.8.3 GUIDED INSTRUCTION *(15 MIN.)*

Learn how to relate equations to contexts

10.8.4 GUIDED PRACTICE *(5 MIN.)*

Practice relating equations to their contexts

10.8.5 TRY IT *(5 MIN.)*

Independent practice relating equations to their contexts

10.8.6 WRAP-UP *(~5 MIN.)*

Reflect on mastery of lesson learning targets

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Calculators
- (Optional) Chart paper and markers
- (Optional) Cups to demonstrate for question 2a in 10.8.2.

ADDITIONAL PREPARATION

- Consider creating and displaying an anchor chart with the properties of equality from previous lessons to aid students in solving two-step equations.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may add or subtract the coefficient of the variable instead of dividing both sides of the equation by the coefficient.
- Students may not distribute to both terms inside of the parentheses.
- Students may not perform the inverse operations on both sides of the equation.
- Students may struggle relating the solution to the real-world context.

THINGS TO CONSIDER

- Model underlining key information in real-world problems. Then, explain how to relate the key information to the equation.
- Ask students to identify something that varies and something that remains constant in the real world.
- Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 10.8.2 Collaboration students match equations to a bar diagram and real-world contexts. Ask students to initially complete the problems independently and then discuss their work. These discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In 10.8.5 Try It, students solve two similar equations and determine which equation represents a real-world problem. Students are asked to explain what the solution to the equation that correctly models this situation means in the context. Ask students whether the solutions make sense given the context. Solving real-world problems increases student interest and appreciation of the utility of mathematics.

DIFFERENTIATE INSTRUCTION

For 10.8.4 Guided Practice, consider allowing students to work with a partner. Students will benefit from hearing each other's reasoning and checking each other's work. These discussions will also provide auditory and visual entry points as students hear and see each other's strategies.

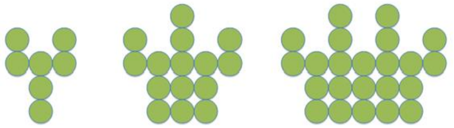
10.8.1 WARM-UP: NUMBER PATTERNS

QUESTIONING

Ask students to define a variable and generalize the pattern with an algebraic expression. Provide students with additional visual patterns and/or ask the student to create their own visual patterns.

10.8.1 Warm-Up: Number Patterns

A pattern is shown.



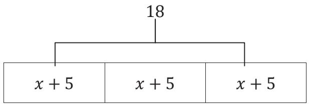
1. Describe how the pattern is changing in each step.
   Answers will vary. In each step you are adding a column of 3 circles and a column of 5 circles to the interior.

2. How many circles will be in the next step?
   31 circles

10.8.2 Collaboration: Matching Equations to Diagrams and Contexts

Equations can be represented using bar diagrams.

1. A bar diagram is shown.



a. Select all the equations that match the diagram.

- ☐ $x+5=18$
- ☐ $18 \div 3 = x+5$
- ☐ $3(x+5)=18$
- ☐ $x+5=\frac{1}{3}\cdot 18$
- ☐ $3x+5=18$

b. Compare your selections with your partner. Resolve any differences if necessary.
Answers will vary.

c. With your partner, choose one of the equations that can be modeled by the bar diagram and explain why it matches the diagram.
Answers will vary.
$3(x+5)=18$ *this is equivalent because there are* 3 *groups of* $x+5$ *which altogether equal* 18.

d. Explain why $x+5=18$ does not match the bar diagram.
$x+5=18$ *does not match the bar diagram because the bar diagram show* 3 *groups of* $x+5$*, not* 1 *group of* $x+5$.

10.8.2 COLLABORATION: MATCHING EQUATIONS TO DIAGRAMS AND CONTEXTS

ENRICHMENT

- What do you notice about the equations that represent the bar diagram?
- What do you notice about the equations that do not match the bar diagram?
- How many x+5 expressions are there?

10.8.2 COLLABORATION: MATCHING EQUATIONS TO DIAGRAMS AND CONTEXTS (CONTINUED)

QUESTIONING

Check that all students understand the contexts in the problems. For example, students may need clarification on the meaning of "nested paper cups." Consider showing students images of the contexts to provide a visual entry point. You can also modify the contexts to appeal to your students' interests.

DIFFERENTIATE INSTRUCTION

Consider splitting students into groups and assigning each group one of the equations (jigsaw strategy). Ask students to show their work on chart paper, including the equation, word problem, and justification of why they match. Use a gallery walk to debrief the problems.

TEACHER MOVES

Before Guided Instruction, review student responses to question 2b as a class. Consider asking these questions:

- What does the variable represent in each equation?
- Will the value of the variable stay the same or change?
- How did you determine which equation matched which context?

Equations can be used to represent real-world contexts.

2. Three equations are shown.

Equation A	Equation B	Equation C
$\frac{1}{4} + 4x = 8$	$4 + \frac{1}{4}x = 8$	$8x + \frac{1}{4} = 4$

a. Work with your partner to determine which equation matches each context. Then, write each equation in the appropriate row.

Context	Equation
A stack of nested paper cups is 8 inches tall. The first cup is 4 inches tall and each of the rest of the cups in the stack adds $\frac{1}{4}$ inch to the height of the stack.	*Equation B*
A baker uses 4 cups of flour. She uses $\frac{1}{4}$ cup to cover the counter with flour so the dough doesn't stick, and the rest to make 8 identical loaves of bread.	*Equation C*
Elena has an 8-foot piece of ribbon. She cuts off a piece that is $\frac{1}{4}$ of a foot long and cuts the remainder into four pieces of equal length.	*Equation A*

b. With your partner, discuss what the variable x represents in each of the equations. Then, write the meaning of x in each equation based on the context it matches.

Equation	Meaning of x
A: $\frac{1}{4} + 4x = 8$	*x is the remaining length of each piece of ribbon after cutting off $\frac{1}{4}$ of a foot.*
B: $4 + \frac{1}{4}x = 8$	*x is the remaining number of cups after the first cup.*
C: $8x + \frac{1}{4} = 4$	*x is the amount of flour in each loaf of bread.*

10.8.3 Guided Instruction: Relating Equations to Their Contexts

Olivia had her birthday party at a trampoline park. It cost $\$48.50$ to reserve a private jumping area for their group and $\$15.72$ per person for food, drinks, and unlimited jump time. Olivia's parents paid $\$315.74$ for Olivia and her friends to attend the party.

1. What was the total amount of money spent on the party?
   $\$315.74$

2. One of the costs depends on the number of people attending the party, while the other does not.

   a. Which cost depends on the number of people who attend the party?
   The cost of the food, drink, and unlimited jump time depends on the number of people at the party.

   b. Which cost is the same regardless of how many people attend the party?
   The cost of reserving the jumping area stays the same regardless of how many people attend the party.

The total cost for Olivia's birthday party can be represented by the equation $315.74 = 15.72n + 48.50$.

3. Work with your partner to answer the following questions.

   a. What does n represent in the equation?
   n represents the number of people attending the party.

   b. What does the variable term, $15.72n$, represent in the equation?
   $15.72n$ represents the cost for food, drinks and unlimited jump time for each person attending the party.

   c. What does the constant term, 48.50, represent in the equation?
   48.50 represents the cost to reserve the jump area.

4. Solve the equation $315.74 = 15.72n + 48.50$.
   $315.74 = 15.72n + 48.50$
   $315.74 - 48.50 = 15.72n + 48.50 - 48.50$
   $267.24 = 15.72n$
   $\frac{267.24}{15.72} = \frac{15.72n}{15.72}$
   $n = 17$

5. If Olivia was included as one of the people attending the party, determine how many of her friends attended.
   16 friends attended the party.

10.8.3 GUIDED INSTRUCTION: RELATING EQUATIONS TO THEIR CONTEXTS

TEACHER MOVES

Engage students with a discussion about the difference between a variable and a constant. Explain that variables are the quantities that change, while constants are the quantities that do not change. Provide a few examples that students can relate to and then ask students to share their own examples.

TEACHER MOVES

After students have answered question 3 with their partner, have students share their responses. Then ask students to relate their answers to question 2 to the equation 315.74=15.72n+48.50.

Why is 15.72 being multiplied by n and 48.5 is not?

10.8.4 GUIDED PRACTICE: RELATING EQUATIONS TO THEIR CONTEXTS

DIFFERENTIATE INSTRUCTION

Consider allowing students to work with a partner. Students may benefit from hearing each other's reasoning and checking each other's work. These discussions will also provide auditory and visual entry-points as students hear and see each other's strategies.

ENRICHMENT

- What part of the equation will stay the same?
- What part of the equation will change?
- Does the solution make sense in the context?

10.8.4 Guided Practice: Relating Equations to Their Contexts

1. Franco is saving money to take a trip to Cocoa Beach, Florida with his friends. Every week, he sets aside the same portion of his paycheck after taking out the $\$325$ he needs for his weekly expenses. After 11 weeks, Franco has $\$489.94$ saved for his trip.

   a. An equation can be created relating these quantities as described below.

   $$\left(\begin{array}{c}\text{Number}\\\text{of}\\\text{weeks}\end{array}\right)\left(\begin{array}{c}\text{Amount}\\\text{of}\\\text{weekly}\\\text{paycheck}\end{array} - \begin{array}{c}\text{Amount}\\\text{of weekly}\\\text{expenses}\end{array}\right) = \begin{array}{c}\text{Total}\\\text{Saved}\end{array}$$

   Which of these quantities is unknown?
   The amount of Franco's weekly paycheck is unknown.

   b. Write the equation that represents this situation using the variable p for the unknown quantity.
   $11(p - 325) = 489.94$

   c. What does $(p - 325)$ represent in the equation?
   The amount that Franco saves each week.

   d. Solve the equation.
   $\frac{11(p-325)}{11} = \frac{489.94}{11}$
   $p - 325 = 44.54$
   $p - 325 + 325 = 44.54 + 325$
   $p = 369.54$

   e. Explain what the solution means in this context.
   Franco earns a paycheck of $\$369.54$ each week.

10.8.5 TRY IT: RELATING EQUATIONS TO THEIR CONTEXTS

TEACHER MOVES

Ask students whether their solution makes sense given the context.

QUESTIONING

Solving real-world problems increases student interest and appreciation of the utility of mathematics. Consider extending the activity by challenging students to create real-world problems that can be represented by equations. Then, students can exchange real-world problems and write the related equations.

10.8.5 Try It: Relating Equations to Their Contexts

Work with your partner to complete the following.

1. Two equations are shown.

   a. Solve both equations. Show your work.

Equation A	Equation B
$46.92 = 6(y - 1.75)$	$6y - 1.75 = 46.92$
$\frac{46.92}{6} = \frac{6(y-1.75)}{6}$	$6y - 1.75 + 1.75 = 46.92 + 1.75$
$7.82 = y - 1.75$	$6y = 48.67$
$7.82 + 1.75 = y - 1.75 + 1.75$	$y \approx 8.11$
$9.57 = y$	

   b. A family of 6 is going to the local fair. They have a coupon for $\$1.75$ off of each ticket. They pay $\$46.92$ for all their tickets. Explain which equation represents this situation.
   Equation A represents this situation because the family is buying 6 tickets but receiving 1.75 off of each ticket, not just one ticket.

   c. Explain what the solution to the equation that correctly models this situation means.
   y represents the price of an individual ticket before the coupon is applied. So, each ticket cost $\$9.57$ before the coupon.

10.8.6 Wrap-Up: Reflection

1. Circle the value on the emoji scale that best represents your understanding of today's lesson.

Answers will vary.

2. Briefly explain your choice.
Answers will vary.

10.8.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

For students who struggle with expressing themselves in writing, consider providing alternative options, such as using an audio recorder, speech-to-text software, or referring to problems in the lesson where they wrote and solved real-world two-step equations.